

Inflation

High-inflation environments make analyzing and forecasting companies' financial performance a challenge. Inflation distorts the financial statements, adding to the difficulty of year-to-year historical comparisons, ratio analyses, and performance forecasts.

When inflation is high, analysis and valuation depend on insights from both nominal- and real-terms approaches. Sometimes nominal indicators are not useful (e.g., for capital turnover). In other cases, real indicators are problematic (e.g., when determining corporate income taxes). But when properly applied, valuations in real and nominal terms should yield an identical value.

Although all the familiar tools described in Part Two still apply to periods of high inflation, such times cause particular complications. This chapter discusses the following issues:

- How inflation leads to lower value creation in companies, because it erodes real-terms free cash flow (FCF), as companies don't increase prices enough to overcome higher capital costs as well as operating costs
- How to evaluate a company's historical performance when inflation is high
- How to prepare financial projections of a company's performance in both nominal and real terms

INFLATION LEADS TO LOWER VALUE CREATION

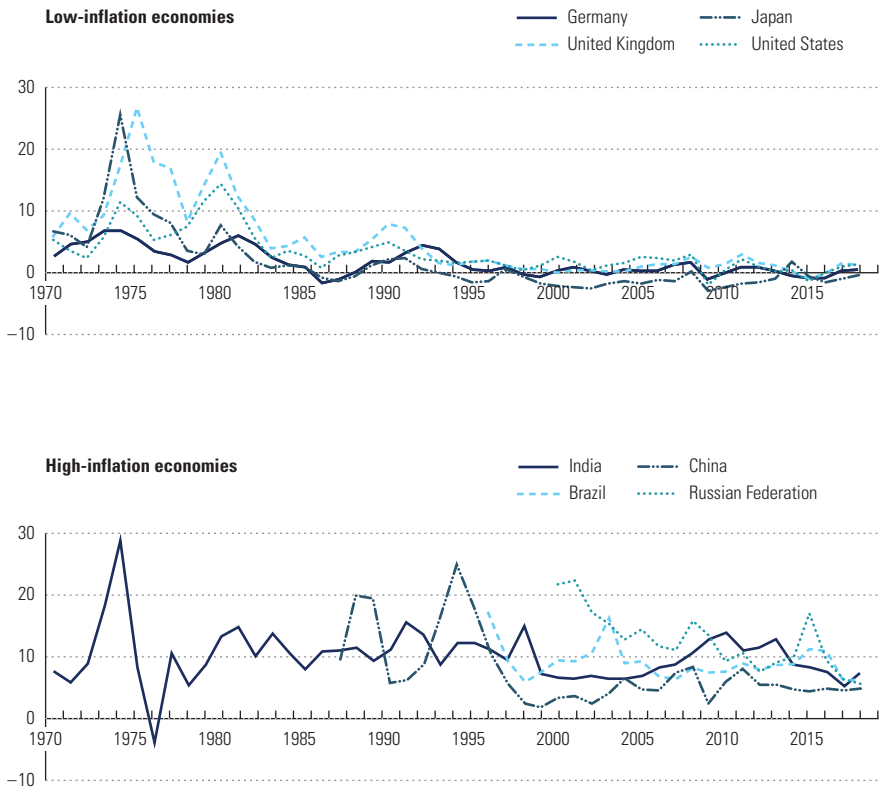
Since the 1980s, inflation has generally been mild in the developed economies of Europe and North America, at levels around 2 to 3 percent per year. But this does not mean inflation has become irrelevant. As Exhibit 26.1 shows, the situation was quite different in the 1970s, when inflation hovered around

10 percent for the same economies. A return to such levels is unlikely, but some economists are warning of rising inflation in, for example, the United States as a result of rising wage levels, import tariffs, and government deficits.¹ And some of the largest economies in Latin America and Asia—such as, for example, Brazil, China, and India—as well as Russia have faced inflation at double-digit levels for intervals of many years. In stark contrast, Japan has experienced extremely low inflation and even deflation since the early 1990s.

Inflation often persists, stretching over several years as it did during the 1970s and 1980s, because suppressing it requires strict and unpopular government measures. For example, curbing inflation caused by overheating in the economy typically requires increasing interest rates and reducing public spending to dampen growth. In most cases, such measures are undertaken

EXHIBIT 26.1 **Historical Inflation Rate in Developed and Emerging Economies**

Annual CPI-based inflation rate, %



Source: *World Development Indicators*, The World Bank. Creative Commons Attribution 4.0 International license (CC-BY 4.0).

¹ See, for example, J. Lahart, "Get Ready to Worry about Inflation Again," *Wall Street Journal*, December 12, 2018; or M. Feldstein, "The Fed Must Reassure Markets on Inflation," *Financial Times*, June 28, 2009.

only when everything else has failed and when inflation has become too high to ignore—but even more difficult to fix.

It's necessary to take account of persistent inflation in analysis and valuation, because a large body of academic research clearly shows that inflation is negatively correlated with stock market returns.² To illustrate, as inflation increased from around 2 or 3 percent in the late 1960s to around 10 percent in the second half of the 1970s, the average price-to-earnings (P/E) ratio for companies in the United States declined from around 18 to below 10. When inflation finally came down, from 1985 onward, P/Es returned to their historical levels.

Inflation has obvious pernicious effects on value creation. Academic research has found evidence that investors often misjudge inflation, which pushes up the cost of capital in real terms and depresses market valuations.³ Inflation creates a one-off loss in value for companies with so-called net monetary assets—that is, asset positions that are fixed in nominal terms.⁴ For example, a balance of receivables loses 10 percent in value when inflation unexpectedly increases by 10 percent. The reverse holds for net monetary liabilities, such as fixed-rate debt. Depending on the relative size of a particular company's receivables, payables, and debt, the direct effect could be positive or negative. Companies also can end up paying higher taxes if their depreciation tax shields are not inflation adjusted for tax purposes—and this is typically the case.

Inflation's most value-destroying impact is not obvious. Though companies may increase prices, most cannot or do not increase them enough to cover both their higher operating costs (salaries and purchased goods) and the higher cost of future capital expenditures. As a result, they fail to maintain profitability in real terms.

To understand how significant the challenge of passing on cost increases can be, consider this simple example. Assume a company generates steady sales of \$1,000 per year. Earnings before interest, taxes, and amortization (EBITA) are \$100, and invested capital is \$1,000. Assume the asset base is evenly spread across 15 groups with remaining lifetimes of 1 to 15 years. Gross property, plant, and equipment (PP&E) is \$1,875, and annual capital expenditures equal depreciation charges at \$125.⁵ The company's key financials would

² See, for example, E. Fama and G. Schwert, "Asset Returns and Inflation," *Journal of Financial Economics* 5 (1977): 115–146; and J. Ritter and R. Warr, "The Decline of Inflation and the Bull Market of 1982–1999," *Journal of Financial and Quantitative Analysis* 37, no. 1 (2002): 29–61.

³ See, for example, F. Modigliani and R. Cohn, "Inflation, Rational Valuation, and the Market," *Financial Analysts Journal* 35 (1979): 24–44; and Ritter and Warr, "The Decline of Inflation," who found that in times of high inflation, investors tend to capitalize real cash flows at nominal discount rates.

⁴ See, for example, H. Hong, "Inflation and the Market Value of the Firm: Theory and Test," *Journal of Finance* 32, no. 4 (1977): 1031–1048.

⁵ At the end of each year, after replacement of the asset group that is fully depreciated, the average remaining life of assets is exactly eight years. Annual depreciation is therefore $\$1,000 \div 8 = \125 , and gross PP&E equals $15 \times \$125 = \$1,875$.

EXHIBIT 26.2 Financial Projections without Inflation

\$	Year 1	Year 2	Year 3	Year 4	Year 16	Year 17
Sales	1,000	1,000	1,000	1,000	1,000	1,000
EBITDA ¹	225	225	225	225	225	225
Depreciation	(125)	(125)	(125)	(125)	(125)	(125)
EBITA²	100	100	100	100	100	100
Gross property, plant, and equipment	1,875	1,875	1,875	1,875	1,875	1,875
Cumulative depreciation	(875)	(875)	(875)	(875)	(875)	(875)
Invested capital	1,000	1,000	1,000	1,000	1,000	1,000
EBITDA	225	225	225	225	225	225
Capital expenditures	(125)	(125)	(125)	(125)	(125)	(125)
Free cash flow (FCF)	100	100	100	100	100	100
EBITA growth, %	—	—	—	—	—	—
EBITA/sales, %	10.0	10.0	10.0	10.0	10.0	10.0
Return on invested capital, %	10.0	10.0	10.0	10.0	10.0	10.0
FCF growth, %		—	—	—	—	—

¹ Earnings before interest, taxes, depreciation, and amortization² Earnings before interest, taxes, and amortization

be as shown in Exhibit 26.2. If the cost of capital is 8 percent, the discounted-cash-flow (DCF) value at the start of year 2—or any year—equals:

$$\text{DCF} = \frac{\$100}{(8\% - 0\%)} = \$1,250$$

Now assume that in year 2, inflation suddenly increases to 15 percent and stays at that level in perpetuity, affecting costs and capital expenditures equally. Let's assume the company increases prices enough that its EBITA grows with inflation and its sales margin (EBITA divided by sales) stays near 10 percent while keeping sales volume and physical production capacity constant. In the process, the company even succeeds in lifting its return on invested capital (ROIC) to almost 20 percent after 15 years (see Exhibit 26.3).

Although these results may be impressive at first sight, a closer inspection of the financial performance reveals significant value destruction. Even though EBITA grows at 15 percent per year, growth in earnings before interest, taxes, depreciation, and amortization (EBITDA) is at only 7 to 8 percent per year because depreciation is recorded at historical nominal cost. As a result, capital spending should exceed depreciation charges to keep physical capacity constant, leading to an actual decline in free cash flow (FCF) in the first few years. FCF growth only gradually rises to the rate of inflation in year 17.⁶

⁶ Given our assumption of an asset lifetime of 15 years, FCF growth gradually increases from 0 to 15 percent until year 17, when a new steady state is reached if inflation remains constant.

EXHIBIT 26.3 Financial Projections with Incomplete Inflation Pass-On

\$	Year 1	Year 2	Year 3	Year 4	Year 16	Year 17
Sales	1,000	1,131	1,283	1,460	7,516	8,644
EBITDA	225	240	259	281	1,210	1,392
Depreciation	(125)	(125)	(126)	(129)	(397)	(456)
EBITA	100	115	132	152	814	936
Gross property, plant, and equipment	1,875	1,894	1,934	1,999	6,840	7,866
Cumulative depreciation	(875)	(875)	(876)	(880)	(2,082)	(2,394)
Invested capital	1,000	1,019	1,058	1,119	4,758	5,472
EBITDA	225	240	259	281	1,210	1,392
Capital expenditures	(125)	(144)	(165)	(190)	(1,017)	(1,170)
Free cash flow (FCF)	100	96	93	91	193	222
EBITA growth, %	—	15.0	15.0	15.0	15.0	15.0
EBITA/sales, %	10.0	10.2	10.3	10.4	10.8	10.8
Return on invested capital, %	10.0	11.5	13.0	14.4	19.7	19.7
FCF growth, %	0.0	−3.7	−3.2	−2.4	14.3	15.0

Combine this with a cost of capital increase to 24 percent,⁷ and the company's value plummets. An explicit DCF valuation with continuing value estimated as of year 17 would show the value at the start of year 2 being as low as \$481.

EXHIBIT 26.4 Financial Projections with Full Inflation Pass-On

\$	Year 1	Year 2	Year 3	Year 4	Year 16	Year 17
Sales	1,000	1,150	1,323	1,521	8,137	9,358
EBITDA	225	259	298	342	1,831	2,105
Depreciation	(125)	(125)	(126)	(129)	(397)	(456)
EBITA	100	134	171	213	1,434	1,649
Gross property, plant, and equipment	1,875	1,894	1,934	1,999	6,840	7,866
Cumulative depreciation	(875)	(875)	(876)	(880)	(2,082)	(2,394)
Invested capital	1,000	1,019	1,058	1,119	4,758	5,472
EBITDA	225	259	298	342	1,831	2,105
Capital expenditures	(125)	(144)	(165)	(190)	(1,017)	(1,170)
Free cash flow (FCF)	100	115	132	152	814	936
EBITA growth, %	—	33.7	28.1	24.5	15.1	15.0
EBITA/sales, %	10.0	11.6	13.0	14.0	17.6	17.6
Return on invested capital, %	10.0	13.4	16.8	20.2	34.7	34.7
FCF growth, %		15.0	15.0	15.0	15.0	15.0

⁷ With inflation at 15 percent, the cost of capital increases from 8 percent to $(1 + 8\%) \times (1 + 15\%) - 1 = 24\%$.

To pass on inflation to customers in full without losing sales volume, the company must increase its *cash flows*, not its earnings, at 15 percent per year (see Exhibit 26.4). In this case, the DCF value at the start of year 2 is fully preserved:

$$\text{DCF} = \frac{\$115}{(24\% - 15\%)} = \$1,250$$

But having all cash flows grow with inflation means that earnings must increase much faster than inflation. As the summary financials show, EBITA growth is now more than 33 percent in year 2. In the same year, the sales margin increases from 10.0 percent to 11.6 percent, and ROIC increases from 10.0 percent to 13.4 percent. After 15 years of constant inflation, the sales margin and ROIC would end up at 17.6 percent and 34.7 percent, respectively. ROIC needs to rise this far to keep up with inflation and the higher cost of capital.⁸

Although this example is stylized, the conclusion applies to all companies: after each acceleration in inflation, we should expect reported earnings to outpace inflation, and reported sales margin and ROIC to increase—even though, in real terms, nothing has changed. Unfortunately, history shows that in periods of inflation, companies do not achieve such big improvements in reported return on invested capital. ROICs remained in the range of 7 to 12 percent in the United States during the 1970s and 1980s, when inflation was at 10 percent or more. If companies had succeeded in passing on inflation effects, they should have reported much higher ROICs in those years. Instead, they hardly managed to keep returns at preinflation levels.

One likely cause is that companies cannot pass on the cost increases to customers without losing volume, or they can pass on increases only with some time lag. Another reason could be that managers do not sufficiently adjust targets for growth of earnings and sales margin when faced with inflation. If a company keeps its sales margins and ROIC constant in times of inflation, cash flows and value are eroding in real terms. Maintaining EBITA growth in line with inflation is also insufficient to sustain a company's value; this is even more the case for a leveraged indicator such as earnings per share.

Whatever the exact reason, history shows that companies do not manage to pass on inflation in full. As a result, their cash flow in real terms declines. In addition, there is empirical evidence that in times of inflation, investors are likely to undervalue stocks as they misjudge inflation's effects.⁹ Lower cash flow and higher cost of capital form a proven recipe for lower share prices, just as occurred in the 1970s and 1980s.

⁸ The reason is that invested capital and depreciation do not grow with inflation immediately. For example, in year 2, annual capital expenditures increase by 15 percent, but this adds only $15\% \times \$125 = \18.75 to invested capital. Assets are acquired at the end of each year and depreciated for the first time in the next year. Annual depreciation changes in year 3 by only a small amount: $1/15 \times 19 = 1.25$. In each year, the company replaces only 1/15 of assets at inflated prices, so it takes 15 years of constant inflation to reach a steady state where capital and depreciation grow at the rate of inflation. As the example shows, sales margin and ROIC increase each year until the steady state in year 17.

⁹ Modigliani and Cohn, "Inflation, Rational Valuation, and the Market"; Ritter and Warr, "The Decline of Inflation."

HISTORICAL ANALYSIS IN TIMES OF HIGH INFLATION

In countries experiencing extreme inflation (more than 25 percent per year), companies often report in year-end currency. In the income statement, items such as revenues and costs that were booked throughout the year are restated at year-end purchasing power. Otherwise, the addition of these items would have no relevance. The balance sheet usually has adjustments to fixed assets, inventory, and equity; the accounts payable and receivables are already in year-end terms.

In most countries, however, financial statements are not adjusted to reflect the effects of inflation. High inflation leads to distortions in the balance sheet and income statement. In the balance sheet, nonmonetary assets, such as inventories and PP&E, are shown at values far below current replacement value. In the income statement, depreciation charges are too low relative to current replacement costs. Sales and costs in December and January of the same year are typically added as if they represented the same purchasing power.

As a result, many financial indicators typically used in historical analyses can be distorted when calculated directly from the financial statements in high-inflation economies. In such circumstances, companies often index their internal management accounts to overcome these issues. If they do not, or if you are conducting an outside-in analysis, at least correct for the following distortions:

- Growth is overstated in times of inflation, so restate it in real terms by deflating with an annual inflation index if sales are evenly spread across the year. If sales are not spread evenly, use quarterly or monthly inflation indexes to deflate the sales in each corresponding interval.
- Capital turnover is typically overstated because operating assets are carried at historical costs. You can approximate the current costs of long-lived assets by adjusting their reported values with an inflation index for their estimated average lifetimes. Or consider developing ratios of real sales relative to physical-capacity indicators appropriate for the sector—for example, sales per square meter in consumer retail. Inventory levels also need restating if turnover is low and inflation is very high.
- Operating margins (operating profit divided by sales) can be overstated because depreciation is too low and slow-moving inventories make large nominal holding gains. Corrections for depreciation charges follow from adjustments to PP&E. You can estimate cash operating expenses at current-cost basis by inflating the reported costs for the average time held in inventory. Alternatively, use historical EBITDA-to-sales ratios to assess the company's performance relative to peers; these ratios at least do not suffer from any depreciation-induced bias.

- Credit ratios and other indicators of capital structure health become distorted and require cautious interpretation. Distortions are especially significant in solvency ratios such as debt to equity or total assets, because long-lived assets are understated relative to replacement costs, and floating-rate debt is expressed in current currency units. As Chapter 33 advises, use coverage ratios such as EBITDA to interest expense.¹⁰ These are less exposed to accounting distortions, because depreciation has no impact on them and debt financing is mostly at floating rates or in foreign currency when inflation is persistent.

FINANCIAL PROJECTIONS IN REAL AND NOMINAL TERMS

When you make financial projections of income statements and balance sheets for a valuation in a high-inflation environment, keep in mind that accounting adjustments should not affect free cash flow. Projections are typically made in either nominal or real terms, but high-inflation environments require a hybrid approach because each single approach has different strengths, as Exhibit 26.5 shows. On the one hand, projecting in real terms makes it difficult to calculate taxes correctly, as tax charges are often based on nominal financial statements. Furthermore, you need to project explicitly the effects of working-capital changes on cash flow, because these do not automatically follow from the annual change in real-terms working capital. On the other hand, using nominal cash flows makes future capital expenditures difficult to project, because the typically stable relationship between revenues and fixed assets does not hold in times of high inflation. This means it will also be difficult to project depreciation charges and EBITA.

EXHIBIT 26.5 **Combining Real and Nominal Approaches to Financial Modeling**

✓✓ Preferred Application

Estimates	Modeling approach	
	Real	Nominal
Operational performance		
Sales	✓✓	✓
EBITDA	✓✓	✓
EBITA	✓✓	—
Capital expenditures	✓✓	—
Investments in working capital	✓✓ ¹	✓
Other		
Income taxes	—	✓✓
Financial statements	✓ ²	✓✓
Continuing value	✓✓ ¹	✓✓

¹ If inflation impact on investments in working capital is explicitly included.

² If inflation corrections are separately modeled and included in income statement and balance sheet.

¹⁰ Distortions occur in the ratio of EBITA to interest coverage if operating profit is overstated due to low depreciation charges and low costs of procured materials.

EXHIBIT 26.6 DCF under Inflation: Operational and Financial Assumptions

	Forecasts						
	Year 1	Year 2	Year 3	Year 4	Year 5	...	Year 25
Operational assumptions							
Real growth rate, %		2	2	2	2	...	2
Real revenues, \$	1,000	1,020	1,040	1,061	1,082	...	1,608
Real EBITDA, \$	300	306	312	318	325	...	483
Net working capital/revenues, %		20	20	20	20	...	20
Real net PP&E/real revenues, %		40	40	40	40	...	40
Lifetime of net PP&E, years	5						
Financial assumptions							
Inflation rate, %		20	10	10	10	...	10
Inflation index	1.00	1.20	1.32	1.45	1.60	...	10.75
Tax rate, %		35	35	35	35	...	35
Real WACC, %		8	8	8	8	...	8
Nominal WACC, %		29.6	18.8	18.8	18.8	...	18.8

To prepare consistent financial projections, you therefore need to use elements of both nominal and real forecasts. This section illustrates how to combine the two approaches in a DCF valuation. The example considers a company whose revenues grow at 2 percent in real terms while the annual inflation rate is 20 percent in the first forecast year and 10 percent thereafter (see Exhibit 26.6). To simplify, we assume that all cash flows occur at the end of the year.¹¹

In practice, financial projections for high-inflation valuations raise many more issues than in this simplified example. Nevertheless, the example is useful for showing how to address some key issues when developing a cash flow forecast in periods of inflation. Using the following step-by-step approach leads to the real and nominal valuation results shown in Exhibit 26.7.

Step 1: Forecast Operating Performance in Real Terms

To the extent possible, convert historical nominal balance sheets and income statements into real terms (usually at the current year's currency value). At a minimum, make a real-terms approximation of the historical development of the key value drivers—growth and return on capital—and the underlying capital turnover and EBITA margin, so you can understand the true

¹¹ At extremely high, fluctuating levels of inflation, however, this assumption could distort financial projections, because the cash flows that accumulate throughout the year are subject to different inflation rates. In such cases, split the year into quarterly or even monthly intervals, project cash flows for each interval, and discount the cash flows at the appropriate discount rate for that interval.

EXHIBIT 26.7 **DCF under Inflation: Real and Nominal Models**

	Nominal projections						Real projections							
	Year 1	Year 2	Year 3	Year 4	Year 5	Year 25	Year 1	Year 2	Year 3	Year 4	Year 5	Year 25		
NOPAT, \$ million														
Revenues	1,000	1,224	1,373	1,541	1,729	...	17,283	1,000	1,020	1,040	1,061	1,082	...	1,608
EBITDA	300	367	412	462	519	...	5,185	300	306	312	318	325	...	483
Depreciation	(80)	(80)	(85)	(92)	(100)	...	(926)	(80)	(80)	(82)	(83)	(85)	...	(126)
EBIT	220	287	327	370	419	...	4,259	220	226	231	235	240	...	356
Taxes	(77)	(101)	(114)	(130)	(147)	...	(1,491)	(77)	(84)	(87)	(89)	(92)	...	(139)
NOPAT ¹	143	187	212	241	272	...	2,768	143	142	144	146	148	...	218
Free cash flow, \$ million														
NOPAT	143	187	212	241	272	...	2,768	143	142	144	146	148	...	218
Depreciation	80	80	85	92	100	...	926	80	80	82	83	85	...	126
Capital expenditures	(80)	(106)	(118)	(133)	(149)	...	(1,491)	(80)	(88)	(90)	(92)	(93)	...	(139)
Investment in net working capital	(45)	(30)	(34)	(38)	(38)	...	(376)	(37)	(37)	(23)	(23)	(24)	...	(35)
Free cash flow	116	149	149	166	185	...	1,827	97	97	113	114	116	...	170
Invested capital, \$ million														
Net PP&E (beginning of year)	400	400	426	459	500	...	4,631	400	400	408	416	424	...	631
Depreciation	(80)	(80)	(85)	(92)	(100)	...	(926)	(80)	(80)	(82)	(83)	(85)	...	(126)
Capital expenditures	80	106	118	133	149	...	1,491	80	88	90	92	93	...	139
Net PP&E (end of year)	400	426	459	500	549	...	5,196	400	408	416	424	433	...	643
Net working capital	200	245	275	308	346	...	3,457	200	204	208	212	216	...	322
Invested Capital	600	670	734	808	895	...	8,653	600	612	624	637	649	...	965
Ratios, %														
Net PP&E/revenues	35	33	33	32	32	...	30	40	40	40	40	40	...	40
Net working capital/revenues	20	20	20	20	20	...	20	20	20	20	20	20	...	20
ROIC	31	32	32	33	34	...	36	24	24	24	23	23	...	23
FCF growth rate		28	11	12	12	...	12		17	1	1	1	...	2
DCF valuation, \$ million														
Free cash flow	116	149	149	166	185	...	1,827	97	97	113	114	116	...	170
Continuing value (Value driver formula) ²							31,063							2,891
Continuing value (Cash flow perpetuity formula)							31,064							2,891
Present value factor	0.77	0.65	0.65	0.55	0.46	...	0.01	0.93	0.93	0.86	0.79	0.74	...	0.16
DCF value, \$ million	1,795							1,795						

¹ Net operating profit after taxes.

² Adjusted formula for real-terms continuing value.

economics of the business. With these approximations, forecast the operating performance of the business in real terms:

- Project future revenues and cash expenses to obtain EBITDA forecasts.¹²
- Estimate PP&E and capital expenditures from your assumptions for real-terms capital turnover.
- Working capital follows from projected revenues and assumptions about days of working capital required.
- From projected net PP&E and assumptions about the lifetime of the assets, derive the annual depreciation to estimate real-terms EBITA.

Step 2: Build Financial Statements in Nominal Terms

Nominal projections can be readily derived through the following steps, which convert the real operating projections into nominal terms:¹³

- Project nominal revenues, cash expenses, EBITDA, and capital expenditures by multiplying their real-terms equivalents by an estimated inflation index for the year.
- Estimate net PP&E on a year-by-year basis from the prior-year balance plus nominal capital expenditures minus nominal depreciation (which is estimated as a percentage of net PP&E according to the estimated asset lifetime).
- Project working capital by multiplying the real-terms amounts by the inflation index for the year (or derive from real-terms revenues and days of working capital required).
- Subtract the nominal depreciation charges from EBITDA to obtain nominal EBITA.
- Calculate income taxes on nominal EBITA without inflation corrections, unless tax laws allow for such corrections.

This example did not build a complete balance sheet and income statement. Complete financial statements would be needed for major decisions concerning, for example, dividend policy and capital structure, debt financing,

¹² This step assumes that all expenses included in EBITDA are cash costs.

¹³ As noted, these projections are made for valuation purposes and not necessarily in accordance with local or international accounting standards prescribing any inflation or monetary corrections for particular groups of assets and liabilities under, for example, inflation accounting. Free cash flows would not be affected by such adjustments.

and share repurchase. Developing complete nominal financial statements would require the following additional steps:

- Forecast interest expense and other nonoperating income statement items in nominal terms (based on the previous year's balance sheet).
- Check that equity equals last year's equity plus earnings, less dividends, plus or minus any share issues or repurchases.
- Balance the balance sheet with debt or marketable securities.

Step 3: Build Financial Statements in Real Terms

Most of the operating items for the real-terms income statement and balance sheet were already estimated in step 1. Now include the real-terms taxes on EBITA by deflating the nominal taxes as estimated in step 2. For full financial statements, use the inflation index to convert debt, marketable securities, interest expense, income taxes, and nonoperating terms from the nominal statements into real terms. The real-terms equity account is a plug to balance the balance sheet. To make sure you have done this correctly, be sure the real equity account equals last year's equity plus earnings, less dividends, plus or minus share issues or repurchases, and plus or minus inflationary gains or losses on the monetary assets and liabilities (such as cash, receivables, payables, and debt).

Step 4: Forecast Free Cash Flows in Real and Nominal Terms

Forecast the future free cash flows in real and nominal terms from the projected income statements and balance sheets. Follow the general approach described in Chapter 10. The only difference is that the real-terms investment in net working capital (NWC^R) is equal to the increase in working capital plus a monetary loss due to inflation:¹⁴

$$\text{Investment in } NWC_t^R = \text{Increase in } NWC_t^R + NWC_{t-1}^R \left(\frac{i_t}{1+i_t} \right)$$

where i_t is the inflation rate in year t .

To check for consistency, use the inflation index to convert the free cash flows from the nominal projections to real terms. These should equal the free cash flows from the real-terms projections in each year.

¹⁴ Even for assets held at constant levels in real-terms balance sheets, replacement investments are required at increasing prices in an inflationary environment. These replacement investments represent a cash outflow, also in real terms, but do not show up from real-terms balance sheet differences from year to year. In contrast, the nominal investment cash flow does follow from the nominal balance sheet differences from year to year.

Step 5: Estimate DCF Value in Real and Nominal Terms

When discounting real and nominal cash flows under high inflation, you must address three key issues:

1. Ensure that the weighted average cost of capital estimates in real terms ($WACC^R$) and nominal terms ($WACC^N$) are defined consistently with the assumptions for inflation (i) in each year:

$$1+WACC_t^N=(1+WACC_t^R)(1+i_t)$$

2. Make sure the explicit forecast period is long enough for the model to reach a steady state with constant growth rates of free cash flow in the year when you apply the continuing-value formula. Because of the way inflation affects capital expenditures and depreciation, you need a much longer horizon than for valuations with no or low inflation.
3. The value driver formula as presented in Chapter 14 can be readily applied when estimating continuing value in nominal terms, but it should be adjusted when estimating in real terms in high-inflation environments. The return on capital in real-terms projections ($ROIC^R$) overestimates the economic returns in the case of positive net working capital. The free cash flow in real terms differs from the cash flow implied by the value driver formula by an amount equal to the annual monetary loss on net working capital:

$$FCF_t^R=\left(1-\frac{g_t^R}{ROIC_t^R}\right)NOPAT_t^R-NWC_{t-1}^R\left(\frac{i_t}{1+i_t}\right)$$

where g^R is growth rate in real terms, and $NOPAT^R$ is net operating profit after taxes in real terms. The real-terms value driver formula is adjusted for this monetary loss, reflecting the perpetuity assumptions for inflation (i) and the ratio of net working capital to invested capital (NWC^R/IC^R):

$$CV^R=\frac{\left(1-\frac{G^R}{ROIC^R}\right)NOPAT^R}{WACC^R-g^R}$$

where

$$G^R=g^R+\left[\frac{NWC^R}{IC^R}\left(\frac{i}{1+i}\right)\right]$$

The resulting continuing-value estimate is the same as that obtained from an FCF perpetuity growth formula. After indexing for inflation, it also equals the continuing-value estimates derived from nominal projections.

Of course, the DCF valuations in nominal and real terms should lead to exactly the same result. Combining both approaches not only provides additional insights into a company's economics under inflation but also is a useful cross-check on the validity of the valuation outcomes.

SUMMARY

High and persistent inflation destroys value because companies typically cannot increase prices enough to offset higher capital outlays. To analyze and value companies in the presence of such inflation, we use the same tools and approaches as introduced in Part Two. However, applying them can be somewhat different.

When analyzing a company's historical performance, you should be aware that persistent inflation can distort many familiar financial indicators, such as growth, capital turnover, operating margins, and solvency ratios. Ensure that you make appropriate adjustments to these ratios. When making financial projections, use a combined nominal- and real-terms approach, because real-terms and nominal-terms projections offer relevant insights and can be used for cross-checking your results. When discounting cash flows, use inflation assumptions in the weighted average cost of capital that are fully consistent with those underlying your cash flow projections.

REVIEW QUESTIONS

1. Why does high inflation typically destroy value for companies?
2. Which company's ROIC would you expect to go up more in times of inflation: a company with long-lived assets or one with short-lived assets, everything else being equal? Why?
3. Describe the impact of high inflation on the financial statements of a company. What unique challenges does inflation present for analysis of historical performance?
4. Explain how an increase in inflation affects a company's tax shields from depreciation and what would be the resulting impact on the company's value.
5. Why should you construct both real and nominal corporate forecasts when doing a valuation in high-inflation conditions?
6. Describe the five-step approach to combining nominal and real forecasts.
7. Assume that inflation unexpectedly increases by 10 percent. Explain why a company's ROIC then needs to increase by more than 10 percent to preserve its shareholder value.
8. In conditions of high inflation, nonmonetary assets tend to be stated on the balance sheet at values far below their replacement costs. Inventory accounting can further complicate historical analysis for companies in such an environment. Which accounting methodology would better represent the true value of the inventory in periods of high inflation: last in, first out (LIFO) or first in, first out (FIFO)? How would this change in a period of deflation?

